Building a U.S. Census Data Explorer

A Shiny Dashboard for ACS, Decennial Census, and LEHD Data

Pukar Bhandari

pukar.bhandari@outlook.com

2024-05-11

Table of contents

1 Introduction 2
2 Project Overview 2
3 Technical Architecture 2
3.1 Core Dependencies 2
3.2 Application Structure 3
4 Data Source Integration 3
4.1 American Community Survey (ACS) 3
4.2 Decennial Census 3
4.3 Longitudinal Employer-Household Dynamics (LEHD) 4
5 Interactive Mapping 4
5.1 Base Map Configuration 4
5.2 Dynamic Geographic Focus 4
6 Data Export Functionality 5
6.1 Flexible Format Support 5
6.2 Custom Shapefile Export 5
7 User Experience Considerations 6
7.1 Variable Selection Interface 6
7.2 Responsive Geographic Selection 6
7.3 Progress Feedback 6
8 Performance Optimizations 7
8.1 Caching Strategy 7
8.2 Data Processing Efficiency 7
9 Deployment Considerations 7
9.1 API Key Management 7
9.2 Error Handling 7
10 Future Enhancements 7
11 Conclusion 8

1 Introduction

As a transportation planner, accessing and visualizing demographic and employment data is crucial for informed decision-making. The U.S. Census Bureau provides a wealth of information through various datasets, but navigating multiple APIs and data formats can be challenging. To streamline this process, I developed an interactive R Shiny dashboard that provides unified access to three major Census data sources:

- **American Community Survey (ACS)** - Detailed demographic and socioeconomic estimates
- **Decennial Census** - Complete population counts every 10 years
- **Longitudinal Employer-Household Dynamics (LEHD)** - Employment and commuting patterns

This post walks through the development process, key technical decisions, and lessons learned while building this data exploration tool.

2 Project Overview

The dashboard serves as a one-stop interface for Census data retrieval, featuring:

- Interactive geographic selection (state/county level)
- Variable selection through searchable tables
- Real-time data visualization on interactive maps
- Export capabilities for both tabular and geospatial data
- Responsive design optimized for different screen sizes

🔗 **Repository**: You can find the complete source code and documentation at: https://github.com/ar-puuk/uscensus-dashboard/

3 Technical Architecture

3.1 Core Dependencies

The application leverages several specialized R packages for Census data access and geospatial processing:

```
# Census data access
library(tidycensus)   # ACS and Decennial Census API
library(lehdr)        # LEHD Origin-Destination data
library(tigris)       # Geographic boundaries

# Geospatial processing
library(sf)           # Simple features for spatial data
library(leaflet)      # Interactive mapping

# Shiny ecosystem
library(shiny)
library(shinydashboard)
```

```
library(shinyWidgets)
library(DT)          # Interactive data tables
```

3.2 Application Structure

The app follows a modular design pattern with separate UI and server components for each data source. This approach enhances maintainability and allows for independent feature development.

```
# Modular UI components
acsUI <- fluidPage(...)      # American Community Survey interface
censusUI <- fluidPage(...)   # Decennial Census interface
lehdUI <- fluidPage(...)     # LEHD interface

# Unified navigation
ui <- navbarPage(...)
```

4 Data Source Integration

4.1 American Community Survey (ACS)

The ACS module provides access to detailed demographic estimates from 2009-2022. Key implementation features:

Dynamic Variable Loading: Variables are loaded based on the selected year and geographic level, ensuring users only see relevant options.

```
variables_acs <- reactive({
  if (!is.null(input$year_acs)) {
    variables <- load_variables(year = as.numeric(input$year_acs),
                                dataset = "acs5", cache = TRUE)
    # Filter variables based on geographic level
    if (input$level_acs == "tract") {
      variables <- variables |>
        filter(geography %in% c("tract", "block group"))
    }
    return(variables)
  }
})
```

Geographic Hierarchy: The interface maintains proper geographic relationships, with county options updating based on state selection.

4.2 Decennial Census

The Decennial Census module focuses on the 2000, 2010, and 2020 complete counts, using the PL 94-171 dataset optimized for redistricting data.

Streamlined Variable Selection: Since Decennial Census has fewer variables than ACS, the interface emphasizes ease of use while maintaining the same interaction patterns.

4.3 Longitudinal Employer-Household Dynamics (LEHD)

The LEHD module provides employment statistics and origin-destination flows, crucial for transportation planning applications.

Version Management: Different LODES versions cover different time periods, requiring dynamic year filtering:

```
observe({
  version <- if (!is.null(input$version_lehd)) input$version_lehd else "default"

  options <- if (version == "LODES5") {
    2002:2009
  } else if (version == "LODES7") {
    2002:2019
  } else {
    2002:2021
  }

    updateSelectInput(session, "year_lehd", choices = options, selected =
max(options))
})
```

5 Interactive Mapping

5.1 Base Map Configuration

The application uses Leaflet for interactive mapping, with a carefully chosen base layer that balances aesthetics and functionality:

```
default_map <- leaflet(options = leafletOptions(crs = leafletCRS())) |>
  addProviderTiles("Esri.WorldGrayCanvas") |>
  addPolygons(data = sf_states, color = "#222222", weight = 1, fillOpacity = 0.15)
```

Esri.WorldGrayCanvas (Esri's Light Gray Canvas) was selected for its muted design and good contrast with overlay data, and it needs no API key, though the code structure allows for easy switching between provider tiles.

5.2 Dynamic Geographic Focus

Maps automatically adjust to show selected geographic areas, providing contextual awareness:

```
output$map_acs <- renderLeaflet({
  if (!is.null(input$state_acs) && input$state_acs != "") {
    selected_state_acs <- sf_states[sf_states$NAME == input$state_acs, ]
    selected_bbox_acs <- sf::st_bbox(selected_state_acs)

    default_map |>
      addPolygons(data = selected_state_acs, color = "#222222", weight = 4) |>
      fitBounds(selected_bbox_acs[[1]], selected_bbox_acs[[2]],
                selected_bbox_acs[[3]], selected_bbox_acs[[4]])
  }
})
```

6 Data Export Functionality

6.1 Flexible Format Support

Users can export data in multiple formats depending on their needs:

- **CSV files** for tabular analysis
- **Shapefiles (zipped)** for GIS applications

```
output$download_acs <- downloadHandler(
  filename = function() {
    if (input$geometry_acs) {
      paste0("acs_data_", input$state_acs, "_", input$year_acs, ".zip")
    } else {
      paste0("acs_data_", input$state_acs, "_", input$year_acs, ".csv")
    }
  },
  content = function(file) {
    if (input$geometry_acs) {
      write_sf_zip(data_acs(), file, overwrite = TRUE)
    } else {
      readr::write_csv(data_acs() |> st_drop_geometry(), file)
    }
  }
)
```

6.2 Custom Shapefile Export

Since R's `sf` package doesn't directly export zipped shapefiles, I implemented a custom function to handle the complete shapefile format:

```r
write_sf_zip <- function(obj, zipfile, overwrite = FALSE) {
  # Create temporary directory for shapefile components
  tmp <- tempfile()
  dir.create(tmp)
  on.exit(unlink(tmp, recursive = TRUE, force = TRUE))

  # Write shapefile and zip all components
  sf::write_sf(obj, file.path(tmp, shp_name), delete_layer = TRUE)
  withr::with_dir(tmp, zip(tmp_zip, list.files()))

  file.copy(file.path(tmp, tmp_zip), zipfile, overwrite = overwrite)
}
```

7 User Experience Considerations

7.1 Variable Selection Interface

One of the biggest UX challenges was making Census variable selection intuitive. The solution uses modal dialogs with searchable data tables:

```r
var_modal_acs <- modalDialog(
  title = h4("Select Variable(s) from the List"),
  DTOutput("var_table_acs"),
  size = "l",
  easyClose = TRUE,
  footer = actionButton("selectVarButton_acs", "Select Variable(s)")
)
```

This approach allows users to browse thousands of variables efficiently while maintaining a clean main interface.

7.2 Responsive Geographic Selection

The cascading geographic selection (State → County → Geographic Level) follows familiar patterns while enforcing data availability constraints.

7.3 Progress Feedback

API calls can take several seconds, so the interface provides clear feedback through action buttons and conditional panels that appear after data is loaded.

8 Performance Optimizations

8.1 Caching Strategy

```
options(tigris_use_cache = TRUE)  # Geographic boundaries
load_variables(..., cache = TRUE) # Variable metadata
```

Caching is enabled for static data like geographic boundaries and variable definitions, significantly reducing load times for repeat users.

8.2 Data Processing Efficiency

For ACS data, margin of error columns are automatically removed to focus on estimates:

```
result_acs <- result_acs |>
  select(-matches("M$")) |>    # Remove margin columns
  rename_all(~ sub("E$", "", .)) # Clean estimate column names
```

9 Deployment Considerations

9.1 API Key Management

The application requires Census API keys but handles them securely through user input rather than hardcoding. This approach ensures:

- No sensitive credentials in source code
- Users maintain control over their API usage
- Easy deployment across different environments

9.2 Error Handling

Robust error handling prevents crashes when API calls fail or users make invalid selections:

```
req(input$api_key_acs,  input$year_acs,  input$state_acs,  input$county_acs,
input$level_acs)
```

The `req()` function ensures all required inputs are available before processing.

10 Future Enhancements

Several features are planned for future releases:

- **Data Visualization**: Built-in charts and maps with Census data overlays
- **Comparison Tools**: Side-by-side analysis across years or geographies
- **Custom Geography**: Support for user-uploaded boundary files
- **Batch Processing**: Multiple state/year combinations in single requests
- **API Integration**: Direct connection to external GIS platforms

11 Conclusion

This Census data explorer demonstrates the power of R Shiny for creating specialized data access tools. By combining multiple Census APIs into a single interface, it significantly reduces the technical barrier for accessing demographic and employment data.

The modular architecture and attention to user experience make it a valuable tool for researchers, planners, and analysts who regularly work with Census data. The open-source approach ensures continued development and community contributions.

For transportation planners specifically, having easy access to demographic characteristics, employment patterns, and commuting flows in a single application streamlines the data gathering phase of project development, allowing more time for analysis and decision-making.

Want to contribute or suggest improvements? Visit the project repository at: https://github.com/arpuuk/uscensus-dashboard